CS19541 - COMPUTER NETWORKS LAB MANUAL

EXPERIMENT 10

AIM: Design and configure an internetwork using wireless router, DHCP server and internet cloud.

DESCRIPTION:

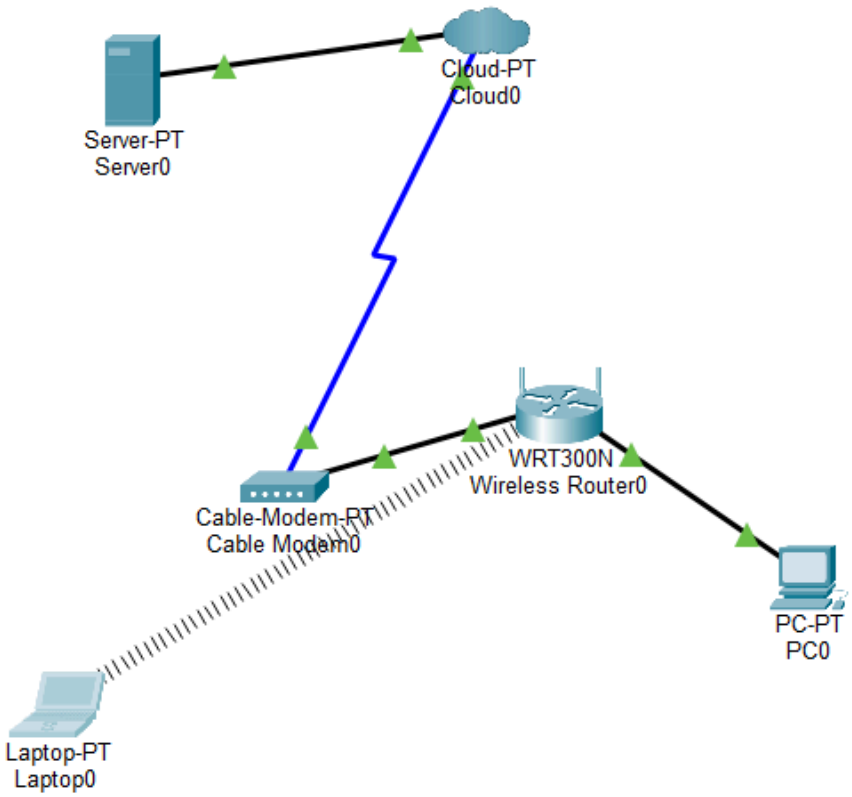
- **Wireless Router & Integrated Services:** A wireless router acts as a gateway combining multiple network functions, including an Access Point (AP), a built-in switch, and a router to connect local wireless and wired clients to external networks like the Internet.
- **Dynamic Host Configuration Protocol (DHCP):** DHCP automatically assigns IP addresses, subnet masks, default gateways, and DNS server details to client devices on the network, simplifying IP management and preventing address conflicts.
- **Internet Cloud Simulation:** Represents the WAN infrastructure connecting the local enterprise or home network to remote servers or services across public routed networks.

STEP-BY-STEP PROCEDURE

Task 1: Topology Setup

1. Place a Wireless Router, DHCP Server, Internet Cloud, and Wireless/Wired Client Devices into the workspace.

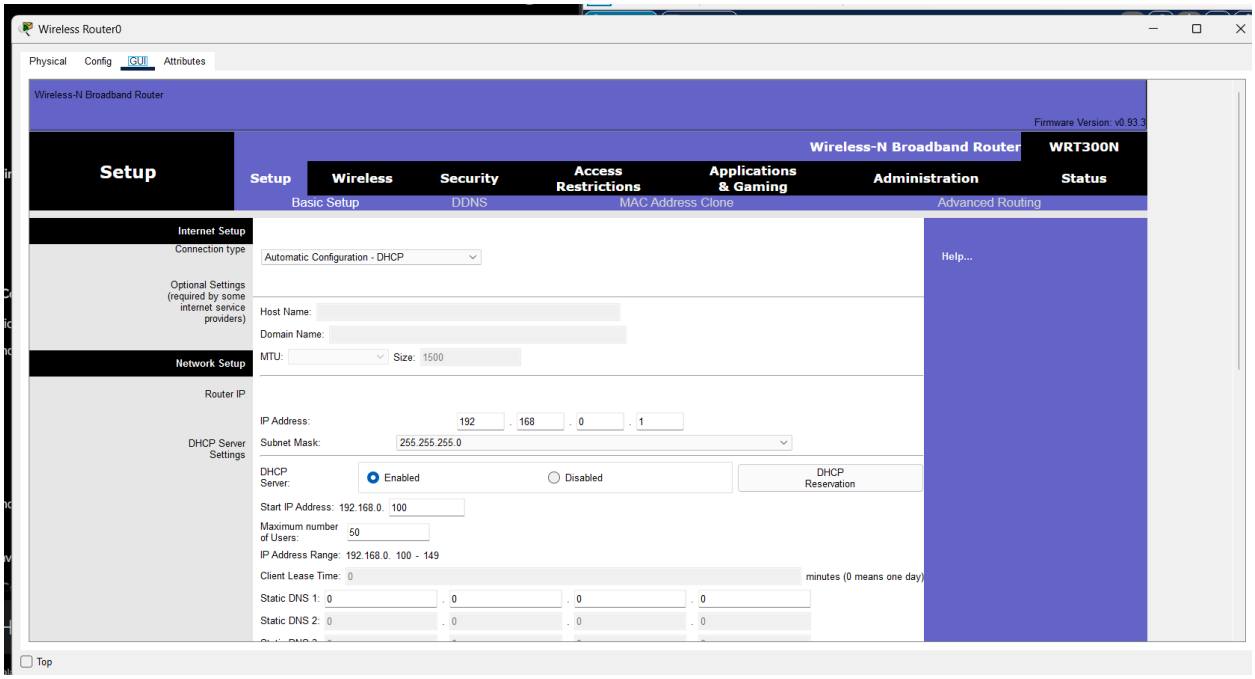
2. Connect the DHCP Server to the Wireless Router using appropriate Ethernet cabling.
3. Connect the Wireless Router's WAN interface to the Internet Cloud.

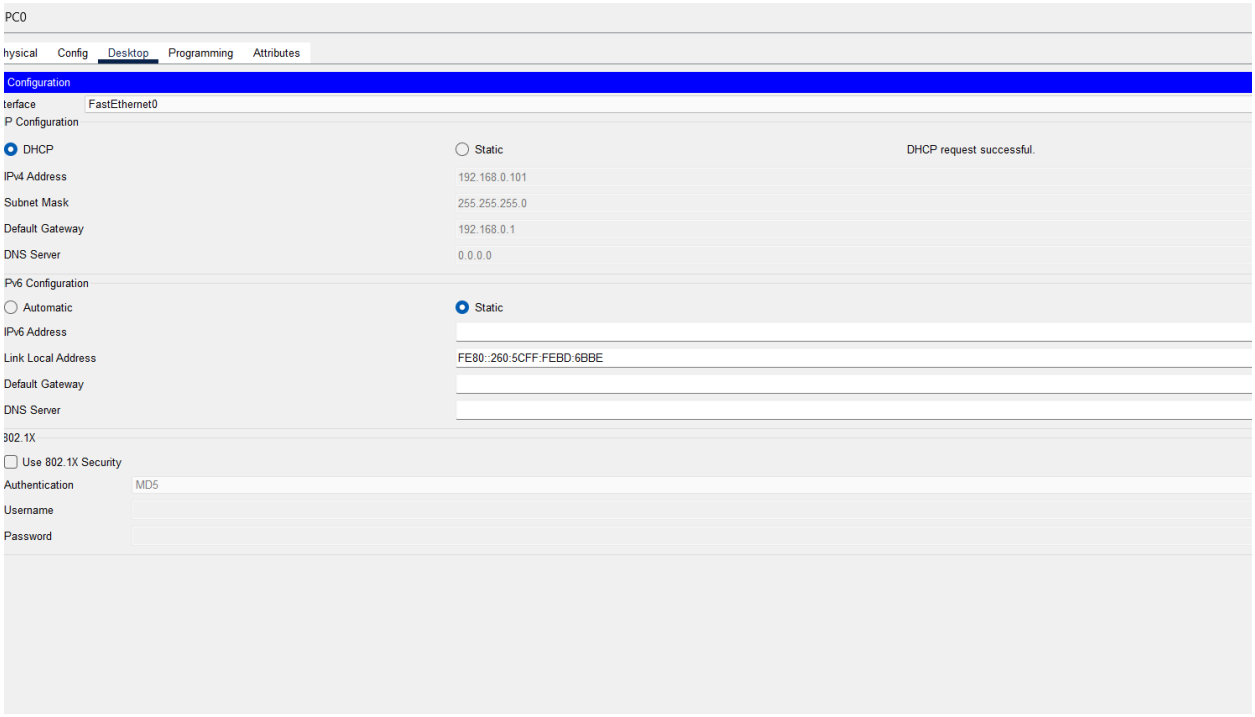


Task 2: DHCP Server & Router Configuration

1. Configure the static IP address and default gateway on the DHCP Server.
2. Enable the DHCP service and set up the IP pool range, default gateway, and DNS settings for clients.

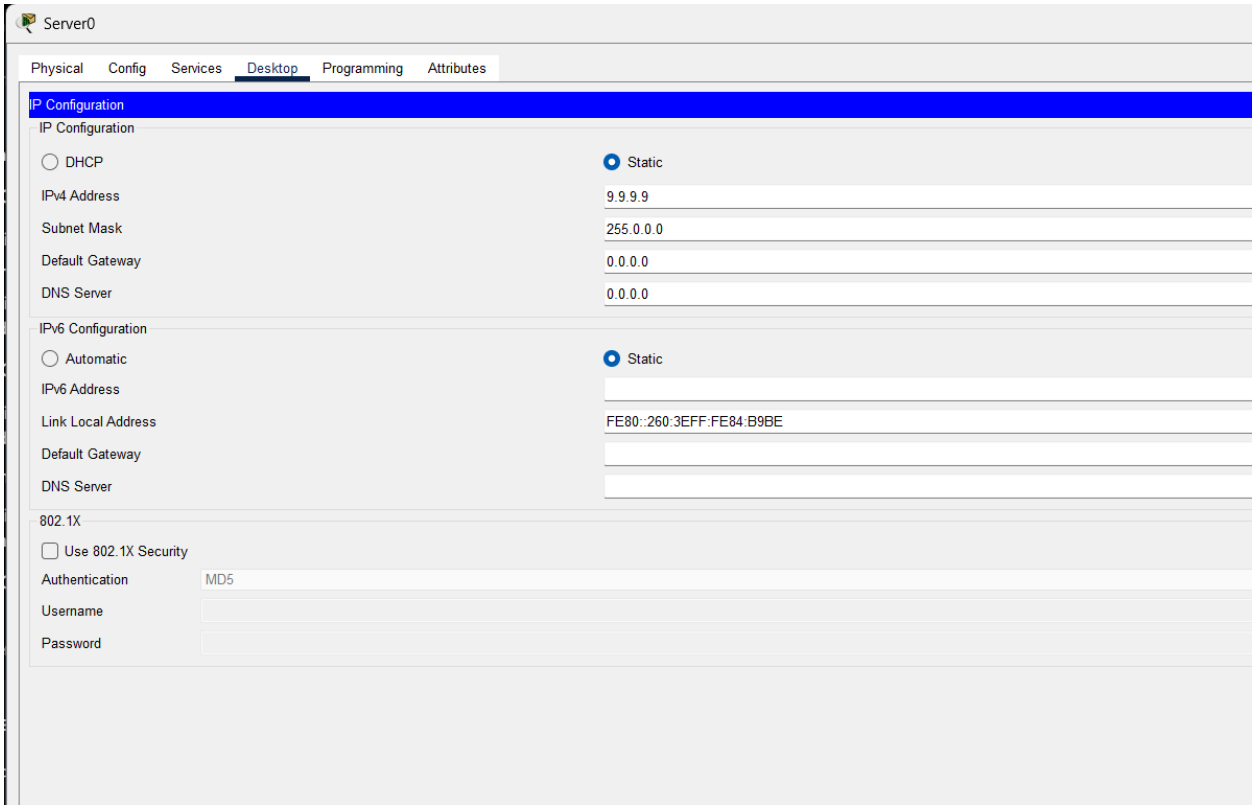
3. Configure wireless SSID, security settings (e.g., WPA2-PSK), and IP routing on the Wireless Router.

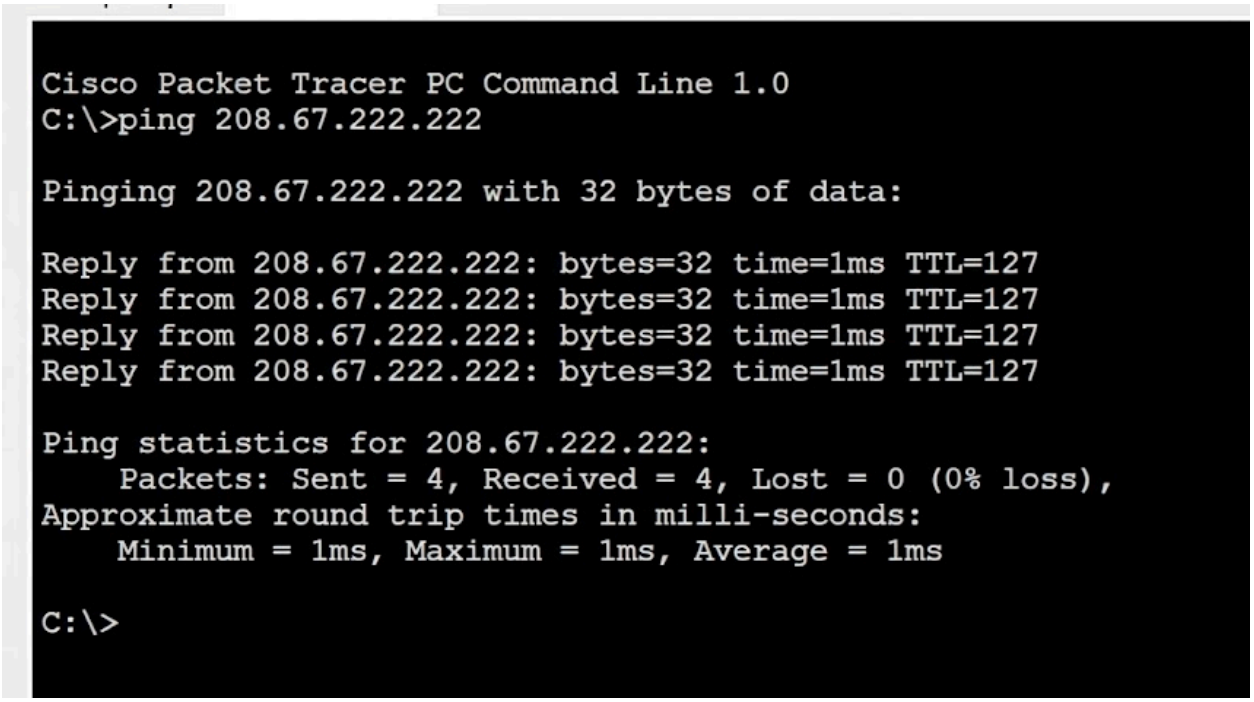




Task 3: Client Verification & Testing

1. Connect client laptops/PCs wirelessly to the router using the configured SSID and security key.
2. Verify that clients dynamically receive IP addresses from the DHCP Server.
3. Test end-to-end connectivity by pinging between local clients, the gateway, and across the internet cloud.





Result:

Thus the experiment to design and configure an internetwork using wireless router, DHCP server and internet cloud has been studied and executed successfully.